RL-Based Smart Streetlight Control with Tabular Q-Learning

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GitHub: `https://github.com/Rohan1729-ju/rl-smart-streetlight`

AI Assignment 2 – Real-Life Use Cases of RL

1 Problem

Streetlights usually run at fixed brightness all night (or all day, with faulty photocells), wasting electricity when roads are empty or already bright. We cast brightness control as a Markov Decision Process in which an agent dims or brightens a lamp every 10 minutes to cut energy use while keeping the road adequately lit when pedestrians are present.

2 RL Formulation

State space S. The observation is discretised into (time-of-day block $\in \{0..5\}$, 4 h each) $\times$ (ambient-light bin $\in \{0..3\}$, thresholds 10/30/60 units) $\times$ (pedestrian bin $\in \{0, 1\text{–}2, 3+\}$) $\times$ (previous brightness level $\in \{0..4\}$), giving $|S| = 6 \cdot 4 \cdot 3 \cdot 5 = 360$ states. Previous brightness makes the flicker penalty Markovian.

Action space A. Five discrete dimming levels $b \in \{0, 0.25, 0.5, 0.75, 1\}$ of a 100 W lamp (60 light units at full output).

Reward R. With ambient light L_a, required light L_{req} (20 units on an empty road, 50 if pedestrians are present) and n pedestrians,

$$r_t = -\alpha\, b_t \;-\; \beta \underbrace{\frac{\max(0, L_{req} - L_a - 60\, b_t)}{L_{req}}}_{\text{shortfall}} (1 + 0.5 \min(n, 4)) \\ -\gamma\, |b_t - b_{t-1}|, \tag{1}$$

with $\alpha = 1$, $\beta = 3$, $\gamma = 0.1$. The first term penalises electricity, the second penalises under-lighting (more heavily when more people are present), and the third discourages flickering.

3 Environment and Algorithm

Simulator (custom, NumPy only). One episode is a 24 h day of 144 steps. Ambient light follows a clipped sinusoid between 06:00 and 18:00 scaled by a per-day cloud factor $\sim U(0.4, 1)$ plus Gaussian noise. Pedestrian counts are Poisson with a time-varying rate with morning (08:00), midday and evening (19:30) peaks and a near-zero night floor. Actions do not affect weather or pedestrians, so every policy can be evaluated on identical days.

Algorithm. Tabular Q-learning, $Q(s,a) \leftarrow Q(s,a) + \eta\,[r + \delta \max_{a'} Q(s',a') - Q(s,a)]$, with $\eta = 0.1$, $\delta = 0.95$, ϵ-greedy exploration decaying from 1.0 to 0.02 ($\times 0.999$/episode), 6000 training episodes. Because all rewards are ≤ 0, a zero-initialised table is optimistic and encourages exploration.

Baselines. Fixed 100%, fixed 50%, uniformly random, and a dusk-to-dawn photocell rule (100% when ambient bin ≤ 1, else off). All are evaluated with the greedy Q-policy on 500 unseen days.

4 Results

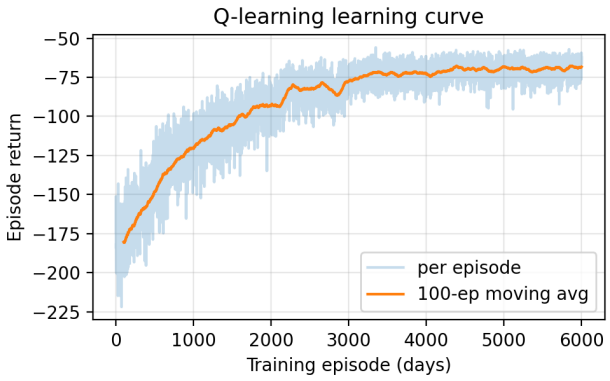


Figure 1: Learning curve: episode return improves from about −180 to −69 and plateaus after ∼4000 episodes.

Policy	kWh/day	Return	Ped. unsafe%
Fixed 100%	2.40	−144.1	0.00
Fixed 50%	1.20	−129.2	50.66
Random	1.20	−186.8	37.58
Dusk-to-dawn	1.57	−103.5	8.21
Q-learning	**1.03**	**−65.6**	**0.08**

Table 1: Mean over 500 unseen days. "Ped. unsafe" = share of pedestrian-present steps with >20% light shortfall.

The agent uses **57% less energy** than fixed 100% brightness (1.03 vs. 2.40 kWh/day) with only 0.08% under-lit pedestrian steps, and beats the dusk-to-dawn rule on both energy (34% lower) and safety. Fixed 50% saves energy but leaves half of pedestrian steps under-lit. Fig. 2 shows the learned policy: off in daylight, about 50% on dark empty roads, and 75–100% when pedestrians are present in the dark. A limitation is flicker: the agent changes level more often (total brightness change 18.5 per day vs. ≤ 1) than fixed policies; raising γ would trade energy for smoothness.

Sensitivity. Sweeping $\beta \in \{0.5, \ldots, 10\}$ (Fig. 3) traces an energy–safety frontier: low β saves energy

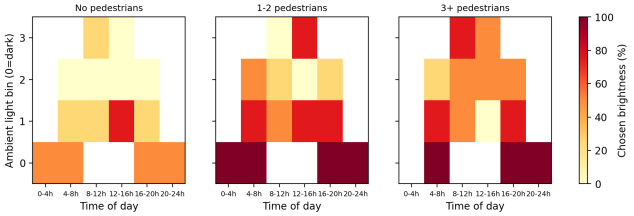


Figure 2: Greedy brightness by time block and ambient bin (previous level 50%). Blank cells are unvisited states.

(0.46 kWh/day) but leaves 51% of pedestrian steps under-lit, while $\beta \geq 2$ reaches near-zero violations at about 1.0–1.1 kWh/day.

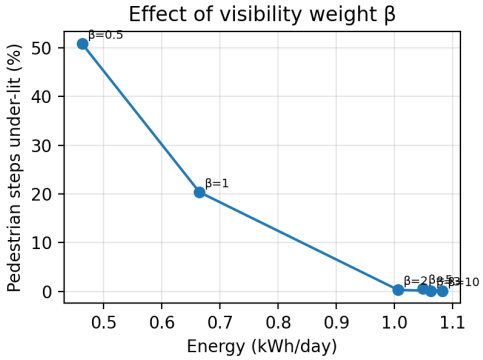


Figure 3: Effect of the visibility weight β on energy and safety.

5 Conclusion

A tabular Q-learning agent learns an interpretable dimming policy that substantially reduces energy while maintaining pedestrian visibility. Future work: multiple coordinated lamps, function approximation (DQN) for continuous states, and real sensor data.